

FEDIOV RESEARCH ANALYSIS

Federated Learning Framework for Vehicular Intrusion Detection

1. Problem Statement & Context

The **Internet of Vehicles (IoV)** landscape is expanding rapidly, but its decentralized and mobile nature makes it highly vulnerable to cyber-attacks like **DoS, message falsification, and data poisoning** [cite: 1281, 1296].

Core Challenges:

- **Privacy vs. Security:** Centralized systems require sharing raw data, compromising user privacy [cite: 1282, 1314].
- **Resource Constraints:** In-vehicle ECUs have limited energy and computation power for heavy models [cite: 1282, 1317].
- **Dynamic Environments:** Non-IID data (different sensor types and driving habits) and high vehicle mobility lead to model instability [cite: 1309, 1315].

2. Methodology: The FedIoV Approach

FedIoV is a secure, adaptive federated framework featuring three major innovations [cite: 1283, 1325]:

- **KANConvNet Classifier:** A lightweight model based on the *Kolmogorov-Arnold theorem*. It uses non-linear activations at the edges of the network, enhancing interpretability and efficiency on ECU-class hardware [cite: 1287, 1328, 1533].
- **Robust Aggregation Pipeline:** A two-stage process using **TOPSIS** for dimensionality reduction (filtering outliers) and **Multi-Krum** for robust clustering (removing malicious/Byzantine updates) [cite: 1286, 1327, 1456].
- **GA-Driven Optimization:** Uses a Genetic Algorithm to automatically tune hyperparameters (e.g., learning rate, batch size) for heterogeneous vehicle fleets [cite: 1288, 1330, 1561].

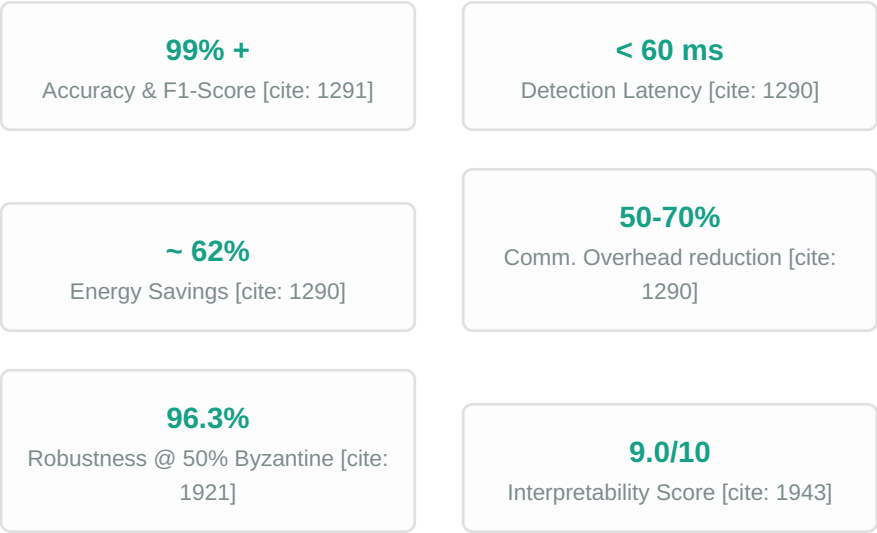
3. Environment & Databases

Simulation Tools: A hybrid testbed utilizing **OMNET++**, **Veins**, and **SUMO** for mobility traces, and **Docker** for containerized vehicle agents [cite: 1654, 1657, 1711].

Dataset	Attack Vectors Covered
Car-Hacking	CAN bus spoofing, fuzzing, and replay [cite: 1678].
VeReMi	V2V position falsification and misbehavior [cite: 1681].
CICIDS 2017	DDoS, brute-force, and general network infiltration [cite: 1685].
IEEE VTC-CAN	DoS and spoofing inside CAN protocols [cite: 1687].
CarNet	Multi-sensor (GPS, LiDAR) and V2X telemetry [cite: 1689].

4. Results & Performance Metrics

FedIoV significantly outperforms state-of-the-art methods like GROW, GPIDS, and BCFL across critical benchmarks [cite: 1289, 1731].



5. Major Limitations

- **Centralized Aggregation:** The central server remains a potential single point of failure and bottleneck in ultra-dense cities [cite: 2420].
- **Computational Scaling:** Multi-Krum aggregation has a quadratic cost $O(N^2)$, which may struggle with thousands of concurrent clients [cite: 1873, 2429].
- **Evolving Threats:** The framework lacks native defense against advanced adaptive backdoors, model inversion, or side-channel attacks [cite: 2422].
- **Benchmark Reliance:** Current validation is heavily based on static benchmarks which may not capture zero-day context-aware threats in the wild [cite: 1672, 2432].

6. Proposed Solutions & Future Directions

The paper proposes several feasible enhancements to transition FedIoV from a prototype to a "road-proven" defense fabric [cite: 2434, 2449]:

- **Hierarchical Aggregation:** Integrating Roadside Units (RSUs) or edge servers to perform localized pre-aggregation before the central server [cite: 1874, 1893].
- **Asynchronous FL:** Replacing synchronous rounds to prevent "stragglers" (delayed vehicles) from halting global model progress [cite: 1895, 2450].
- **Privacy Hardening:** Implementing **Differential Privacy** noise locally or using **Trusted Execution Enclaves (TEEs)** to protect model updates from inversion [cite: 1724, 2423, 2452].
- **Continual Learning:** Adding layers to adapt models across geographic regions without "catastrophic forgetting" of previous safety patterns [cite: 2451].